

Kaiwen (Kevin) He

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Research Interest and Vision

Applied cryptography: I design and implement *efficient* cryptographic solutions for the *security* and *privacy* of everyone.

Education

Massachusetts Institute of Technology	Cambridge, MA
• Ph.D. candidate in Computer Science	Sep 2023 – Current
• M.S. in Computer Science	Sep 2023 – Sep 2025
University of California San Diego	La Jolla, CA
• B.S. in Computer Engineering	Sep 2020 – Jun 2023

Experience

Teaching Assistant , MIT – Cambridge, MA	Jan 2026 – Current
• 6.5610, Applied Cryptography.	
Research Assistant , MIT – Cambridge, MA	Jun 2024 – Jan 2026
• 4.0× acceleration of RSA signing in BoringSSL (used by Chrome and most browsers) in the recent work of VROOM.	
• 45× acceleration of multi-key homomorphic secret sharing with applications to fuzzy password-based key exchange.	
Research Experiences for Undergraduates , UC San Diego – La Jolla, CA	Jun 2022 – May 2023
• Collected weekly data from 2^{32} or 4 billion hosts (the entire IP address space).	
• Designed a new open source ZGrab 2.0 module with 7829 lines of code to collect data from IPsec hosts.	

Talks

Northeastern University , Crypto Reading Group	Boston, MA
<i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i>	March 2026
Boston University , BUSEC Seminar	Boston, MA
<i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i>	February 2026
Google	New York, NY
<i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i>	January 2026
Berkeley Security Seminar	Berkeley, CA
<i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i>	January 2026
UCSD Security Lunch	La Jolla, CA
<i>Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications</i>	January 2026
MIT CIS Seminar (with Lali Devadas)	Cambridge, MA
<i>Multi-Key Homomorphic Secret Sharing, From Theory To Practice</i>	December 2025
CSAW	New York, NY
<i>Passive SSH Key Compromise via Lattices</i>	November 2024
ACM CCS	Copenhagen, Denmark
<i>Passive SSH Key Compromise via Lattices</i>	November 2023

Awards and Honors

Finalist , Qualcomm Innovation Fellowship	March 2026
Most notable paper: technical impact , CSAW Applied Research Competition	November 2024
<i>Passive SSH Key Compromise via Lattices</i>	

Irwin Mark Jacobs and Joan Klein Jacobs Presidential Fellowship, MIT September 2023
Offered to newly admitted Ph.D. students who have demonstrated exemplary academic and research achievements, and thus show great promise for future accomplishments.

SIM San Diego Scholarship, Society of Information Management (SIM) San Diego October 2022
Offered to nominated students only.

Publications

Concretely-Efficient Multi-Key Homomorphic Secret Sharing and Applications May 2026
Kaiwen He, Sacha Servan-Schreiber, Geoffroy Couteau, Srinivas Devadas
IEEE S&P 2026 (to appear)

Passive SSH Key Compromise via Lattices November 2023
Keegan Ryan, Kaiwen He, George Arnold Sullivan, Nadia Heninger
ACM CCS 2023

Critique of: “A Parallel Framework for Constraint-Based Bayesian Network Learning via Markov Blanket Discovery” by SCC Team From UC San Diego October 2022
Arunav Gupta, John Ge, John Li, Zihao Kong, Kaiwen He, Matthew Mikhailov, Bryan Chin, Xiaochen Li, Max Apodaca, Paul Rodriguez, Mahidar Tatineni, Mary Thomas, and Santosh Bhatt
IEEE TPDS 2022

Preprints

VROOM: Accelerating (Almost All) Number-Theoretic Cryptography Using Vectorization and the Residue Number System February 2026
Simon Langowski, Kaiwen He, Srinivas Devadas
<https://eprint.iacr.org/2026/393>

Skills

Programming Languages: Python, JavaScript, Go, Java, Bash, C, C++, Rust, TypeScript, x86 and ARM Assembly, Kotlin.
Other: Cryptography, Cryptanalysis, Technical Writing, Technical Presentation, Research.